

Property Tax Overrides, Municipal Credit Ratings, and Voter Support

Evidence from Massachusetts

Tingli Qu¹, Yuzhu Li², and Gang Chen²

¹ American University

² University at Albany, SUNY

2026-05-19

Background

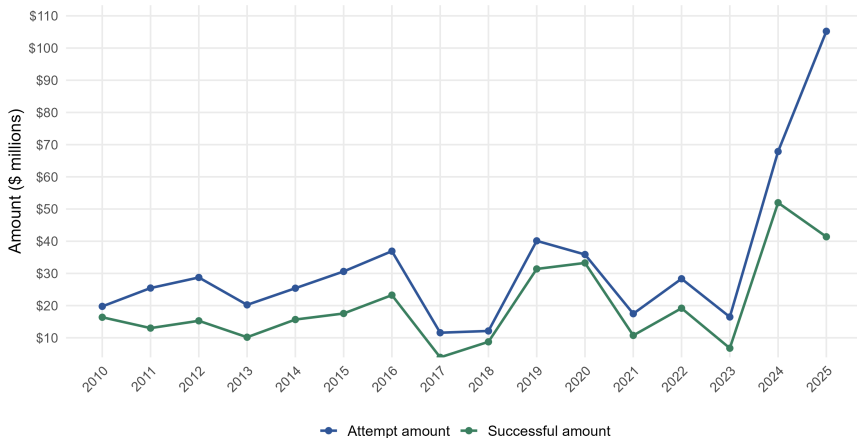
- Driven by concerns over government expansion and increasing tax burdens, **Tax and Expenditure Limits (TELs)** act as institutional constraints on local governments' ability to raise revenues and expenditures (Brennan & Buchanan, 1980).
- TELs commonly restrict property tax growth but often reduce fiscal flexibility needed to respond to shocks (Joyce & Mullins, 1991; Jimenez, 2018).
- With rising inflation, increasing service costs, and uncertainty surrounding federal and state aid, local governments face significant fiscal challenges.
- Municipal governments in Massachusetts use tax overrides as voter-approved mechanisms to allow temporary tax increases above the property tax limits (Bradbury, Mayer, & Case, 2001).
- Overrides represent a collective decision-making process between governments and voters under fiscal constraints.

Massachusetts Proposition 2½

- A Tax and Expenditure Limit passed by Massachusetts voters in 1980.
- Primarily limits local property taxes through two caps:
 - **Levy ceiling:** total taxes cannot exceed 2.5% of a community's total assessed property value.
 - **Levy limit:** annual property tax revenue growth is capped at 2.5%, plus allowances for new growth.
- Includes voter-approved overrides that allow communities to exceed the limits temporarily for specific needs.
- Provides the institutional setting for studying how governments seek voter support for tax increases and how such efforts affect fiscal outcomes.

Override Amounts in Massachusetts Municipalities

Override attempts and successful overrides, 2010-2025



Research Questions

- ① What is the impact of municipal tax overrides on municipal credit ratings?
- ② How do municipal tax overrides influence subsequent voter support for tax increases?

Dual-Signal Framework

- Override actions convey complex, potentially contradictory signals to rating agencies regarding fiscal health and governance capacity.
- A simple “good” or “bad” assumption is insufficient to capture override signals.
- The dual-signal framework explicitly acknowledges this tension:
 - Overrides may signal fiscal flexibility and political capacity.
 - Overrides may also signal fiscal stress or weak fiscal discipline.

Positive Signal of Overrides

- Overrides can signal fiscal strength and effective governance to credit markets.
- Initiatives signal proactive governance and internal consensus even before passage.
- Successful overrides demonstrate enhanced fiscal capacity and political capital, both valued by rating agencies.
- **H1a:** Property tax override initiatives (attempts) positively affect municipal credit ratings.
- **H1b:** Successful tax overrides positively affect municipal credit ratings.

Negative Signal of Overrides

- The need for an override can signal underlying fiscal stress or mismanagement.
- It may reveal revenue insufficiency, a primary indicator of fiscal distress (Jimenez, 2018).
- Observers may perceive override requests as weak fiscal discipline rather than as responses to external pressures.
- Increased perceived default risk is typically penalized by credit markets.
- **H2a:** Property tax override initiatives (attempts) negatively affect municipal credit ratings.
- **H2b:** Failed tax overrides negatively affect municipal credit ratings.

Political Consequences of Frequent Overrides

- Repeated requests can erode trust and increase voter skepticism about government efficiency.
- This may mobilize opposition and lower future override success rates.
- Credit markets assess long-term stability, so persistent tax fatigue may signal heightened political risk.
- Reduced political capacity implies less future fiscal flexibility when overrides are needed.
- **H3a:** Frequent overrides and high total override amounts negatively correlate with citizens' support for future tax increases.
- **H3b:** Frequent overrides negatively correlate with credit ratings.

- **Moody's ordered probit complete cases:** 265 municipalities, 2003-2015, 2017, and 2019-2021.
- **Constructed regression panel:** 351 municipalities, 2003-2021.
- **Repeated-event DiD:** clean override-event years from 2005-2019, stacked panel years from 2003-2021.

Outcomes Variables and Treatments

- **Rating outcome:** Rating_ordered, the ordered Moody's rating score.
- **Voter-support outcome:** yes votes as a percentage of total votes in a municipality-year.
- **Repeated-event rating-change outcomes:**
 - Rating_downgrade
 - Rating_upgrade
- **Override measures:**
 - Attempt, success, and failure indicators for rating models.
 - Annual and three-year cumulative attempt, success, and failure counts for yes-vote-percentage models.
- **Controls:** population size, debt burden, unemployment, revenue stability, revenue per capita, excess levy capacity, fiscal reserves, and budget balance.

Main Rating Model: Ordered Probit with Mundlak Controls

$$Rating_ordered_{it}^* = \alpha_1 Overrides_{it} + X_{it}\gamma + \bar{X}_i\theta + \delta_t + \varepsilon_{it} \quad (1)$$

- Ordered probit maps the latent rating index to the observed Moody's rating categories.
- The Mundlak approach (Greene, 2004; Wooldridge, 2010) includes the time-averaged values of each time-varying predictor for each municipality to partially approximate municipality fixed effects
- Overrides include attempt, success, and failure indicators.
- Standard errors are clustered by municipality.

Voter-Support Model: Two-way Fixed Effects

$$Support_percent_{it} = \alpha_1 Frequency_{it} + X_{it}\gamma + \mu_i + \delta_t + \varepsilon_{it} \quad (2)$$

- Frequency is annual or three-year cumulative override activity (attempts, successes, failures).
- Municipality fixed effects absorb time-invariant local differences.
- Year fixed effects absorb common statewide shocks.
- Standard errors are clustered by municipality.

Repeated-Event DiD Design

$$\begin{aligned} Rating_change_{ist} = & \sum_{h \neq -1} \beta_h 1[rel_year_{ist} = h] \times Treated_{is} \\ & + \eta_{si} + \tau_{st} + \varepsilon_{ist} \end{aligned} \quad (3)$$

- Event types: override attempts, successful overrides, and failed overrides.
- Event window: $h = -2, -1, 0, 1, 2$ with $h = -1$ as the reference year.
- The primary comparison pool uses window-clean controls with no override event within the event window.
- Fixed effects: stack-by-municipality and stack-by-year.
- Standard errors are clustered by municipality.

Repeated-Event DiD Stacks

- Each clean treated event creates one event-centered stack.
- A stack includes one treated municipality-event plus all window-clean control municipalities observed in the same event window.
- Window-clean stacks:
 - Override attempts: 200 stacks; median 201 municipalities per stack.
 - Successful overrides: 176 stacks; median 247 municipalities per stack.
 - Failed overrides: 144 stacks; median 258 municipalities per stack.
- Municipalities can appear in multiple stacks when they are eligible controls for multiple clean events.

Overrides and Moody's Ratings

- Override attempts and successful overrides are positively associated with ordered Moody's ratings.
- Failed overrides have a small, insignificant association in the active ordered probit models.
- These estimates use controls, Mundlak means, year indicators, and municipality-clustered standard errors.

	Attempt	Success	Failure
Override term	0.130*** (0.048)	0.187*** (0.063)	0.026 (0.055)
Observations	3,550	3,550	3,550
Municipalities	265	265	265
Controls, Mundlak means	Yes	Yes	Yes
Year indicators	Yes	Yes	Yes

Ordered probit models for Moody's ordered rating outcome. Standard errors in parentheses are clustered by municipality. Moody's complete-case years are 2003–2015, 2017, and 2019–2021. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Override Frequency and Moody's Ratings

- Three-year cumulative frequency estimates are negative for attempts, successes, and failures.
- The estimates are small and insignificant, so the models do not provide clear support for H3b.
- Estimates use controls, Mundlak means, year indicators, and municipality-clustered standard errors.

	3-year attempt	3-year success	3-year failure
Frequency term	-0.009 (0.025)	-0.014 (0.051)	-0.010 (0.022)
Observations	3,550	3,550	3,550
Municipalities	265	265	265
Controls, Mundlak means	Yes	Yes	Yes
Year indicators	Yes	Yes	Yes

Ordered probit models for Moody's ordered rating outcome using three-year cumulative operating override counts. Standard errors in parentheses are clustered by municipality. Moody's complete-case years are 2003–2015, 2017, and 2019–2021. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Annual Override Frequency and Yes Votes

- More annual operating override attempts are associated with lower yes-vote percentages.
- Annual successful override counts are positively associated with yes-vote percentages.
- Annual failed override counts are negatively associated with yes-vote percentages.
- Models use municipality and year fixed effects.

	Annual attempt	Annual success	Annual failure
Frequency term	-0.657*** (0.189)	2.618*** (0.659)	-2.025*** (0.351)
Observations	730	730	730
Municipalities	221	221	221
Municipality FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Linear fixed-effects models for annual operating yes-vote percentage using annual operating override counts. Standard errors in parentheses are clustered by municipality. Complete-case years are 2003–2021. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Three-Year Override Frequency and Yes Votes

- Three-year cumulative attempts are negative but insignificant.
- Three-year cumulative successes are positively associated with yes-vote percentages.
- Three-year cumulative failures are negatively associated with yes-vote percentages.
- Models use municipality and year fixed effects.

	3-year attempt	3-year success	3-year failure
Frequency term	-0.156 (0.162)	1.066*** (0.239)	-0.875*** (0.241)
Observations	730	730	730
Municipalities	221	221	221
Municipality FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Linear fixed-effects models for annual operating yes-vote percentage using three-year cumulative operating override counts. Standard errors in parentheses are clustered by municipality. Complete-case years are 2003–2021. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Repeated-Event Sample Construction

- Clean treated events have no nearby same-type event inside the local event window.
- The design compares treated events with window-clean controls.
- Event-year estimates compare rating-change outcomes to $h = -1$ within each stack.

Event	Focal yrs	Event munis.	Clean evts	Clean munis.	Ctrl munis.
Attempt	784	226	200	145	333
Success	535	184	176	129	340
Failure	357	160	144	112	350

Clean treated events have no same-type event inside the local event window. The comparison pool is window-clean controls. Event years are 2005–2019; stacked panel years are 2003–2021.

Repeated-Event DiD: Downgrades

- Attempt and success event-year downgrade estimates are positive but insignificant.
- Failed-event downgrade estimates are negative in years $h = 1$ and $h = 2$.
- Estimates are relative to $h = -1$ within each stack.

Event	$h = -2$	$h = 0$	$h = 1$	$h = 2$
Attempt	-0.020 (0.016)	0.025 (0.022)	-0.011 (0.013)	-0.020 (0.015)
Success	-0.035*** (0.010)	0.029 (0.023)	0.019 (0.020)	0.001 (0.021)
Failure	-0.004 (0.016)	0.018 (0.022)	-0.024** (0.012)	-0.029** (0.014)

Repeated-event linear probability models for downgrade outcomes with stack-by-municipality and stack-by-year fixed effects. Coefficients are relative to $h = -1$; standard errors in parentheses are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Repeated-Event DiD: Upgrades

- Upgrade estimates are generally negative for attempts and successful overrides.
- Event-year estimates for attempts and successes are statistically significant.
- Estimates are relative to $h = -1$ within each stack.

Event	$h = -2$	$h = 0$	$h = 1$	$h = 2$
Attempt	-0.017 (0.011)	-0.029** (0.014)	-0.027 (0.019)	-0.013 (0.026)
Success	-0.016 (0.010)	-0.029** (0.015)	-0.036 (0.022)	-0.031 (0.026)
Failure	-0.009 (0.013)	-0.012 (0.020)	-0.015 (0.029)	-0.030 (0.037)

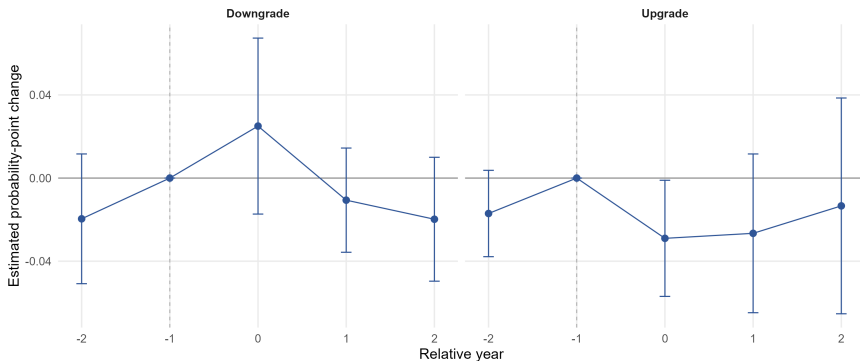
Repeated-event linear probability models for upgrade outcomes with stack-by-municipality and stack-by-year fixed effects. Coefficients are relative to $h = -1$; standard errors in parentheses are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Event Study: Override Attempts

- Upgrade probability falls in the event year; downgrade estimates are insignificant.

Override Attempt

Window-clean repeated-event DiD; coefficients relative to $h = -1$

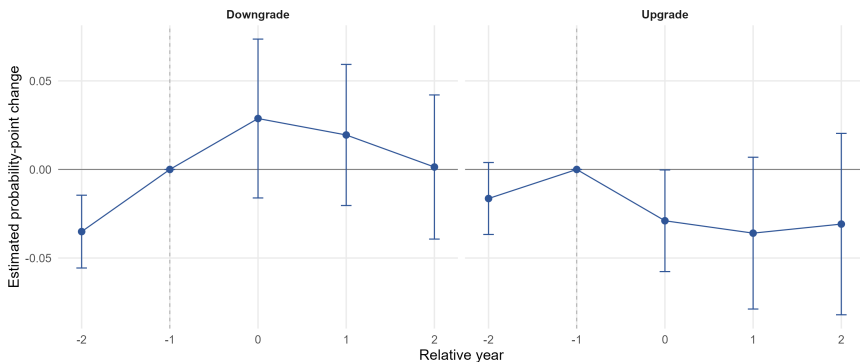


Event Study: Successful Overrides

- Upgrade probability falls in the event year; downgrade estimates remain insignificant.

Successful Overrides

Window-clean repeated-event DiD; coefficients relative to $h = -1$

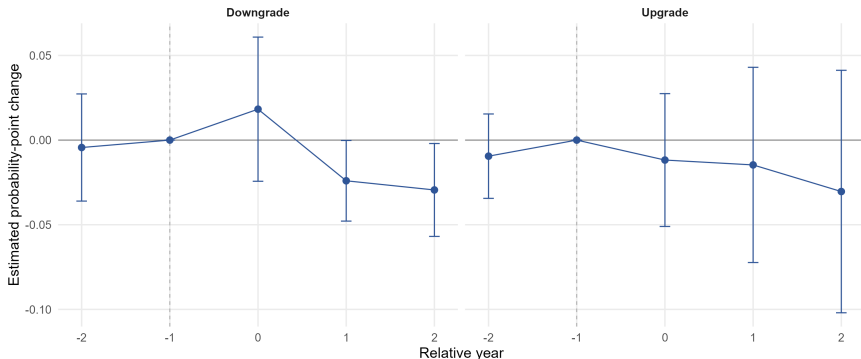


Event Study: Failed Overrides

- Post-event downgrade estimates are negative in $h = 1$ and $h = 2$, suggesting fewer additional downgrades relative to $h = -1$. This may mean that rating agencies may already observe fiscal problems before the failed vote.

Failed Overrides

Window-clean repeated-event DiD; coefficients relative to $h = -1$



Summary of Findings

- Credit rating models show positive associations for attempts and successes, with no clear rating association for failures.
- Voter support models suggest repeated attempts and failures are associated with lower voter support, while successful histories are associated with higher yes-vote percentages.
- Repeated-event DiD models with binary downgrade and upgrade outcomes provide mixed short-window evidence around override events.

Conclusion and Implications

The findings suggest that tax overrides are a double-edged fiscal tool, with potential benefits and costs that vary depending on their frequency and success.

For future research:

- Under what conditions do overrides become a viable strategy for local governments to cope with fiscal challenges and signal fiscal resilience without eroding political support?
- To what extent do override mechanisms reinforce existing fiscal inequalities across communities?
- In the context of declining federal and state aid and rising service costs, how can TELs balance fiscal discipline with necessary flexibility?

Thank You

Thank you! Questions and comments are welcome.

Tingli Qu (tq1939a@american.edu)

Yuzhu Li (yli92@albany.edu)

Gang Chen (gchen3@albany.edu)